

DIFFERENCE BETWEEN JPA, HIBERNATE AND SPRING DATA JPA

INTRODUCTION

JPA (Java Persistence API), Hibernate and Spring Data JPA are technologies used in Java applications to interact with relational databases. They help developers perform database operations such as creating, reading, updating, and deleting data while reducing the amount of SQL code that needs to be written.

JAVA PERSISTENCE API (JPA)

JPA (Java Persistence API) is a specification (JSR 338) that defines a standard way to map Java objects to database tables. It provides guidelines for object-relational mapping (ORM) and database operations but does not contain any implementation. Therefore, JPA requires an implementation provider such as Hibernate to perform actual database operations.

HIBERNATE

Hibernate is an ORM (Object Relational Mapping) framework that implements the JPA specification. It maps Java objects to database tables and automatically generates SQL queries. In Hibernate, developers manually manage sessions and transactions while performing database operations.

SPRING DATA JPA

Spring Data JPA is a Spring Framework module built on top of JPA. It does not implement JPA but simplifies database access by reducing boilerplate code. It provides the `JpaRepository` interface, which includes built-in methods for CRUD operations and automatically manages transactions.

FEATURE	JPA	HIBERNATE	SPRING DATA JPA
Meaning	A specification that defines how persistence should work.	An ORM framework that implements JPA.	A Spring module that makes JPA easier to use.
Type	Specification	Framework	Spring Module

Purpose	Defines rules for mapping Java objects to database tables.	Performs actual database operations.	Makes database operations easier with less code.
Implementation	No implementation.	Provides JPA implementation.	Uses JPA implementation (usually Hibernate).
Database Operations	Cannot perform operations by itself.	Performs CRUD operations.	Performs CRUD operations using JPA repositories.
Boilerplate Code	More code is required.	Requires session and transaction handling.	Very little code is required.
Transaction Management	Not provided.	Managed manually.	Managed automatically using <code>@Transactional</code> .
Repository Support	Not available.	Not available.	Provides <code>JpaRepository</code> for CRUD operations.
Ease of Use	Moderate	Moderate	Easy to use because many tasks are automated.
Common Use	Used as a standard persistence API.	Used as the JPA implementation to interact with databases.	Widely used in Spring Boot applications for rapid development.

The above table is the comparison between JPA, Hibernate and Spring Data JPA.

CONCLUSION

JPA, Hibernate, and Spring Data JPA work together to simplify database operations in Java applications. JPA defines the standard for persistence, Hibernate implements that standard and performs database operations, while Spring Data JPA provides an easier and more efficient way to use JPA by reducing boilerplate code and automatically managing transactions. Together, they make Java database programming simpler, faster, and more maintainable.